MTH:2310 DISCRETE MATH
EXAM 2 REVIEW

- **Recommended Preparation**
  - Re-work examples from class.
  - Work/re-work suggested problems from the book.
  - Re-work quiz problems. Understand and learn from mistakes.
  - Go over this study guide and make sure you know and understand every point on this study guide.
  - Practice showing all your steps. Make solutions organized and be explicit/intentional in your writing.
  - Replicate the exam environment. Can you do relevant problems with little to no help?

1.6 **Rules of Inference**
  - Be able to define the term "valid argument."
  - Be able to identify the argument form of a given argument.
  - Be able to determine and justify whether an argument is valid or invalid.

  Ex. Section 1.6, Exercises 1, 2, 6, 10, 13, 16, 19, 20 **additionally try to fully identify each argument form**

1.7 **Introduction to Proofs**
  - Be able to properly form a direct proof.
    * All variables defined
    * Follows Universal Generalization where applicable
    * Formal conclusion
  - Be able to properly form a proof by contraposition.
    * All variables defined
    * Follows Universal Generalization where applicable
    * Formal conclusion
  - Be able to identify mistakes in a complete proof.

Ex. Section 1.7, Exercises 6*, 8*, 10, 11, 17, 19, 20
Problems marked with $(*)$ *were not originally included in the Suggested Problems list*

2.1 **Sets**

- Be able to define "set."
- Be able to use set-builder notation and roster notation to define a set.
- Be able to determine a set's cardinality.
- Be able to visually represent several sets using a Venn diagram
- Be able to determine whether a given object is an element of set, a subset of a set, or neither.
- Be able to find the power set of a given set.
- Be able to find the truth set of a given propositional function.

Ex. Section 2.1, Exercises 1, 2, 10, 12*, 22*, 23*, 46 *Problems marked with* $(*)$ *were not originally included in the Suggested Problems list*

2.2 **Set Operations**

- Be able to prove that two sets are equal using only set-builder notation.
- Be able to find the union, intersection, and difference of two or more sets.
- Be able to find the complement of a set.
- Be able to draw a Venn diagram which satisfies given relations between sets.
- Be able to shade in portions of a Venn diagram that belong to a specified subset.

Ex. Section 2.2, Exercises 4, 9, 12, 13, 20, 24, 30

Ex. Let A, B, C be non-empty sets and U be the universal set. Suppose $A \cap C \neq \emptyset$, $A \cap B = \emptyset$, $B \subseteq C$, $C \setminus B \neq \emptyset$. Draw a Venn diagram of A, B, C, U. Shade the subset corresponding to $\overline{A \cap C} \setminus \overline{B}$.

2.3 **Functions**

- Be able to determine whether some $f(x)$ defines a function.
- Be able to draw a "bubble-arrow" diagram for a function or composition of functions.
- Be able to define a functions which satisfies given properties.
- Be able to prove a function is injective, surjective, and bijective using the definition.
- Be able to find the composition of two or more functions.

Ex. Section 2.3, Exercises 2, 10 - 13, 20, 35

2.4 **Sequences and Summations** (This may get dropped)

- Be able to find the first N terms of a sequence.
- Be able to completely describe a sequence using words, or as an equation $a_n = f(n)$.
- Be able to show that a sequence is a solution of a given recurrence relation.
- Be able to find a solution of a recurrence relation.
- Be able to evaluate single sums and double sums.
- Be able to use closed forms to evaluate special sums.

Ex. Section 2.4, Exercises 4, 10, 13, 16, 31, 33

Ex. Find a formula for the a_n term of the following sequences

* $\{2, -1, -4, -7, -10, -13, ...\}$
* $\{1, 2, 5, 10, 17, 26, 37, ...\}$
* $\{-3, -6, -12, -24, -48, ...\}$
* $\{1, -1, 2, -6, 24, -120, ...\}$